SRIRAM M

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PROFESSIONALSUMMARY

AI/ML Engineer with hands-on experience in LLM application development, RAG pipelines, and production ML deployment using FastAPI, Docker, and AWS. Proven track record in GenAI systems (LangChain, LangGraph, FAISS), computer vision (YOLOv8, PyTorch), and end-to-end MLOps including model training, evaluation, CI/CD, and cloud deployment. Seeking AI/ML Engineer roles to build scalable, production-grade AI systems. AI/ML Engineer with hands-on experience in LLM application development, RAG pipelines, and production ML deployment using FastAPI, Docker, and AWS. Proven track record in GenAI systems (LangChain, LangGraph, FAISS), computer vision (YOLOv8, PyTorch), and end-to-end MLOps including model training, evaluation, CI/CD, and cloud deployment. Seeking AI/ML Engineer roles to build scalable, production-grade AI systems

.EDUCATION

Master of Computer Applications (MCA) | *Kumaraguru College of Technology, Coimbatore* | 2024 – 2026
GPA: 8.4 / 10

Bachelor of Computer Applications (BCA) | *PSG College of Arts and Science, Coimbatore* | 2021 – 2024 GPA: 7.0 / 10

TECHNICALSKILLS

Programming: Python, SQL
Machine Learning: Scikit-learn, XGBoost, Feature Engineering, Model Evaluation, Hyperparameter Optimization
Deep Learning: PyTorch, TensorFlow, Neural Networks, Computer Vision, NLP
Generative AI: LLM Applications, RAG, LangChain, LangGraph, LlamaIndex, FAISS, Qdrant, Embeddings, Prompt Engineering, AI Agents, LoRA / QLoRA ,Vector Database ,N8n
MLOps & Deployment: FastAPI, Docker, AWS EC2, MLflow, Redis, Celery, CI/CD, Git

PROFESSIONALEXPERIENCE

AI/ML Engineer Intern | *ROR Technologies, Coimbatore (Onsite)* | Dec 2025 – June 2026

- Built and deployed an NLP-based customer support chatbot using FastAPI and LangChain, reducing average query resolution time by ~40% across 500+ daily user interactions.
- Developed a predictive analytics model for business forecasting using XGBoost; achieved AUC-ROC of 0.91 after iterative feature engineering on a 150K-row dataset.
- Designed and implemented REST API endpoints for real-time ML model inference, supporting concurrent requests with sub-200ms response latency using async FastAPI.
- Automated a data preprocessing and model retraining pipeline using Celery and Redis, reducing manual intervention by ~60% and cutting pipeline turnaround time from 4 hours to 45 minutes.
- Containerized AI applications using Docker and deployed to AWS EC2, enabling consistent staging-to-production delivery across 3 internal business use cases.
- Collaborated in code reviews, wrote unit tests, and maintained CI/CD pipelines for ML model versioning and deployment using Git and MLflow.

P R O J E C T E X P E R I E N C E

PROJECT 1 — Enterprise RAG + Agentic Knowledge Assistant

Designed and deployed a production-grade Retrieval-Augmented Generation (RAG) system using LangChain and LangGraph, where an AI agent dynamically routes user queries to either semantic vector search (FAISS, Qdrant) or direct LLM response. Implemented hybrid search combining dense embeddings (Hugging Face Transformers) with BM25 keyword retrieval and cross-encoder re-ranking to improve retrieval precision by 31%. Evaluated pipeline quality end-to-end using RAGAS metrics including faithfulness, answer relevancy, and context precision. Deployed as an async FastAPI microservice with SSE streaming, containerized via Docker, and hosted on AWS EC2 with Redis caching for sub-300ms response latency across 1,000+ document corpus.

PROJECT 2 — Fine-Tuned LLM Microservice (LoRA/QLoRA + Quantized Deployment)

Fine-tuned Llama-3.1-8B on a domain-specific task (SQL generation from natural language) using parameter-efficient fine-tuning (PEFT) with LoRA/QLoRA adapters, reducing trainable parameters by 94% while achieving 88% execution accuracy on held-out test set. Applied 4-bit quantization using bitsandbytes to reduce model size from 16GB to 4.2GB with less than 2% accuracy degradation. Tracked all experiments with MLflow including hyperparameters, loss curves, and evaluation metrics. Exported quantized model to ONNX format and pushed to Hugging Face Hub. Served via async FastAPI batch inference endpoint containerized in Docker, with automated CI/CD pipeline using GitHub Actions for model versioning and deployment.

PROJECT 3 — Real-Time Computer Vision Pipeline + MLOps Monitoring
Built an end-to-end real-time computer vision system using YOLOv8 and PyTorch for industrial defect detection, achieving 93% mAP@0.5 through custom data augmentation, anchor tuning, and transfer learning on a 10,000-image labeled dataset. Deployed inference as a containerized FastAPI service with Celery and Redis for async batch processing, achieving under 45ms per-frame latency. Orchestrated multi-replica deployment using Kubernetes on AWS with zero-downtime rolling updates. Implemented full MLOps loop: experiment tracking with Weights & Biases and MLflow, model versioning and A/B testing between model versions, and live data drift and prediction distribution monitoring using Evidently AI. Automated CI/CD pipeline using GitHub Actions for test, build, push, and deploy stages.

PROJECT 4 — Multi-Agent Autonomous Workflow Platform

Architected a production multi-agent autonomous workflow platform using LangGraph where specialized agents collaboratively handle end-to-end customer support triage: a classifier agent routes incoming requests, a RAG retrieval agent queries a vector knowledge base (Qdrant + Hugging Face embeddings), a response-drafting agent generates replies, and an escalation agent flags edge cases for human-in-the-loop review. Implemented persistent memory, inter-agent communication, and tool-calling (web search, SQL database query, REST API calls) across the agent graph. Integrated n8n webhooks for external workflow triggers (email, Slack, CRM events). Deployed via FastAPI backend with Docker, Redis for state persistence, and a Streamlit monitoring dashboard showing live agent execution logs, resolution rates, and escalation metrics

C E R T I F I C A T I O N S

- AI Coder: Vibe Coder to Agentic Engineer — Udemy, 2025
- Deep Learning & Reinforcement Learning — Coursera, 2024
- Generative AI Models & Platforms — IBM / Coursera, 2024
- Getting Started with DevOps on AWS — AWS Training, 2024
- Python Programming — GeeksforGeeks, 2023
- Prompt Engineering Basics — Coursera, 2024

A D D I T I O N A L I N F O R M A T I O N

Languages: English (Fluent), Tamil (Native), Kannada (Conversational)

Interests: Traveling, Fitness, Coding, MotoGP, Formula 1